

Global Consensus-Based Formation Control of Nonholonomic Mobile Robots With Time-Varying Delays and Without Velocity Measurements

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Abstract—In this article, we propose a solution to the full-consensus-based formation problem for torque-controlled nonholonomic mobile robots under time-varying communication delays and without velocity measurements. Under the assumptions of static and undirected communication topologies, and smooth and bounded delays, we propose a distributed smooth, time-varying control law which achieves the goal of acquiring a global formation pattern in a common consensus point for *both* position and orientation. The controller is a combination of a simple proportional-plus-damping (PD) scheme and a simple globally exponentially convergent velocity observer. Consensus is reached independently of all initial conditions. Realistic simulations in the Gazebo-robot operating system environment illustrate the effectiveness of the proposed algorithm and the robustness with respect to measurement noise, nonsmooth delays, and uncertain dynamics.

Index Terms—Consensus, nonholonomic robots, PD controller, velocity observer.

I. INTRODUCTION

The deployment of fleets of robotic vehicles has been shown to be very effective for a variety of civil and military applications. Mobile robots, in particular, are amply used to achieve cooperative tasks, such as reconnaissance, surveillance, load transportation, and imaging. However, to achieve extensive autonomous deployment of such multirobot systems, it is necessary to provide distributed algorithms whose design is based on realistic assumptions about the vehicle model and the available information obtained by the onboard sensors. Such a design will be more easily implementable and will allow us to guarantee desired performances during deployment.

The starting point for multirobot cooperation control is the distributed consensus problem, which has received considerable attention in the last decades [1]. For fleets of mobile robots, the two main

categories of problems addressed in the literature are partial consensus, in position [1], [2] [3], [4], or only in orientation [5], and full consensus [6], [7]. Position (orientation) consensus refers to the objective of all robots converging to a common position (orientation) with an arbitrary or predetermined orientation. On the other hand, in full consensus, we seek agreement in both position and orientation. Moreover, we talk of formation-based consensus since our goal is to include the physical space occupancy of the robot and as such impose a formation pattern with an unknown center. This is achieved by imposing an offset position from the center of the formation for each robot.

The common practice in the control design for networks of mobile robots is to consider simple models, such as kinematic ones and simple or double integrators [8]. These last ones can also be obtained through dynamic feedback linearization [9] but are not ideal from a robustness viewpoint. Some other works, as [10], deal with the leader–follower formation of nonlinear systems with delays via output-feedback controllers. However, since mobile robots are subject to nonholonomic constraints and are in reality force- and torque-controlled, both kinematic and dynamic models need to be considered. A complete design model is also necessary when velocity information is *unavailable* and needs to be reconstructed to be used in the controller.

Solutions to the problem of consensus of nonholonomic mobile robots have been proposed, for instance, in [6] where a distributed law achieves full consensus, with the controller being discontinuous and time invariant. The feasibility of a class of position formations was studied in [3] where the authors provide necessary and sufficient conditions. In [2], the authors consider the kinematic and dynamic models in position and propose a smooth distributed formation control law using a consensus-based approach. In these references, as well as a large number of the literature, the working assumption is that the communication network is completely reliable and not affected by delays. In addition, these works are based on simplified kinematic models of the nonholonomic mobile robots. A more realistic second-order, torque-controlled dynamics model is used in [11].

These protocols may be implemented using absolute position and orientation values [2], [7] or rather, using relative measurements [12], [13], [14]. In a practical deployment, however, communication delays are omnipresent, affecting the exchanged information between agents, and can significantly deteriorate the cooperative behavior. This presents an additional challenge that has to be explicitly accounted for during the controller synthesis while assuming that each agent has access to its neighbors' information at all times without delays is not realistic. The effect of delays in the coordination of nonholonomic agents has been studied, for instance, in [15] and [16]. However, both consider only constant delays and kinematic models. On the other hand, in spite of the majority of works on control of nonholonomic systems, very few address the position-feedback control problem in a multiagent setting; see, e.g., [17], [18] on the leader–follower problem or [19], [20] on leaderless consensus, using approximate differentiation.

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None of these works deal with interconnecting delays. Except the recent work [21], where a virtual second-order dynamical system is employed, for each robot, to inject the required damping in the system to achieve the consensus-based objective under the presence of communication delays.

The consensus-based formation control of nonholonomic vehicles is a set-point stabilization problem, and thus, this problem cannot be solved via time-invariant smooth feedback [22]. In this regard, the controller has to be either discontinuous or time varying.

In this article, we provide a simple solution for the (leaderless) consensus-based formation control of multiple nonholonomic vehicles under static, undirected communications. We simultaneously consider the effects of the following:

- 1) realistic second-order torque-controlled dynamics;
- 2) position- and orientation-only measurements;
- 3) smooth and bounded but varying, time delays.

We propose a smooth controller of proportional-plus-damping type, which achieves the control goal through a vanishing time-varying term—cf. [23], and a cascade interconnection with an Immersion and Invariance velocity observer [24], which estimates the robot's linear and angular velocities.

The rest of this article is organized as follows. Section II presents the underlying models and the problem formulation. Section III presents the proposed distributed law and the corresponding stability analysis. In Section IV, realistic simulations in the Gazebo/robot operating system (ROS) environment illustrate the effectiveness and robustness of our algorithm. Finally, Section V concludes this article.

II. MODEL AND PROBLEM FORMULATION

Consider a group of N force-controlled nonholonomic vehicles, each of which is modeled by the usual unicycle kinematics equations and two added integrators, that is, for each $i \leq N$, we have

$$\dot{z}_i = \varphi(\theta_i)v_i, \quad \varphi(\theta_i) := [\cos(\theta_i), \sin(\theta_i)]^\top \quad (1a)$$

$$\dot{v}_i = u_{vi}, \quad u_{vi} := \frac{1}{r_i m_i} [\tau_{i1} + \tau_{i2}] \quad (1b)$$

$$\dot{\theta}_i = \omega_i \quad (2a)$$

$$\dot{\omega}_i = u_{\omega i}, \quad u_{\omega i} := \frac{R_i}{I_i r_i} [\tau_{i1} - \tau_{i2}] \quad (2b)$$

where $z_i := [x_i, y_i]^\top \in \mathbb{R}^2$ are the Cartesian coordinates of the robot on the plane, $\theta_i \in \mathbb{R}$ denotes its orientation, v_i and ω_i are the linear and angular velocities of the centre of mass, respectively, m_i is the mass, I_i is the moment of inertia, r_i is the radius of the wheels, R_i is the distance between point z_i and the wheels, u_{vi} and $u_{\omega i}$ are the control inputs, and τ_{ij} are the torques at the wheels, that is, for the purpose of control implementation, we have

$$\begin{bmatrix} \tau_{i1} \\ \tau_{i2} \end{bmatrix} = \frac{r_i}{2} \begin{bmatrix} m_i & I_i/R_i \\ m_i & -I_i/R_i \end{bmatrix} \begin{bmatrix} u_{vi} \\ u_{\omega i} \end{bmatrix}. \quad (3)$$

Our goal is to design u_{vi} and $u_{\omega i}$ to achieve the desired consensus-based formation objective. This objective involves making the robots acquire a desired formation pattern relatively to a barycentre. More precisely, let $\delta_i \in \mathbb{R}^2$ be a constant vector determining the position of the i th vehicle relatively to the barycentre and, correspondingly, let $\tilde{z}_i := z_i - \delta_i$ denote its relative position error. If the coordinates of the barycentre are imposed as a reference set-point, denoted hereafter by z_ℓ , and a reference orientation, θ_ℓ , we speak of leader–follower consensus. Otherwise, if the barycentre's Cartesian coordinates, denoted z_c , and a common orientation, denoted θ_c , result from the properties of the

interconnection graph, they are unknown and we speak of leaderless consensus.

We assume that each robot possesses position sensors and is able to communicate its position to a number of neighbors through a reliable network. That is, we make the following assumptions.

(A1) The interconnection graph is undirected, static, and connected.

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Therefore, the interconnection graph may be modeled by a constant Laplacian matrix, $L := [\ell_{ij}] \in \mathbb{R}^{N \times N}$, where

$$\ell_{ij} = \begin{cases} \sum_{k \in \mathcal{N}_i} a_{ik} & i = j \\ -a_{ij} & i \neq j. \end{cases} \quad (4)$$

$\mathcal{N}_i \subset \mathbb{Z}$ is the set of indices corresponding to robots transmitting information to the i th robot, $a_{ij} > 0$ if $j \in \mathcal{N}_i$ and $a_{ij} = 0$ otherwise. By construction, $L\mathbf{1}_N = 0$, where $\mathbf{1}_N = [1 \cdots 1]^\top$ and, after Assumption A1, L is symmetric, it has a unique zero-eigenvalue, and the rest of its spectrum is strictly positive. Thus, $\text{rank}(L) = N - 1$. Furthermore, it is also assumed that

(A2) the communication from the j th to the i th robot is subject to a variable time-delay denoted $T_{ji}(t)$ that is bounded by a known upper bound $\bar{T}_{ji} \geq 0$ and has bounded time derivatives.

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Then, the control problem consists in designing u_{vi} and $u_{\omega i}$ so that

$$\lim_{t \rightarrow \infty} \tilde{z}_i(t) = z_c; \quad \lim_{t \rightarrow \infty} \theta_i(t) = \theta_c \quad \forall i \leq N \quad (5)$$

$$\lim_{t \rightarrow \infty} v_i(t) = 0; \quad \lim_{t \rightarrow \infty} \omega_i(t) = 0 \quad \forall i \leq N \quad (6)$$

and

$$\lim_{t \rightarrow \infty} \hat{v}_i(t) - v_i(t) = 0; \quad \lim_{t \rightarrow \infty} \hat{\omega}_i(t) - \omega_i(t) = 0 \quad \forall i \leq N$$

where $\hat{v}_i(t)$ and $\hat{\omega}_i(t)$ are the estimators of the linear and angular velocities, of each agent, respectively, and the center of the formation z_c and the agreement orientation θ_c are *unknown* and they result from the interconnection topology and the robots initial positions.

III. MAIN RESULTS

In this section, we present the main results of the note.

A. I & I Velocity Observer

Following the results in [25]—see also [26], [27], in this section, we propose a globally convergent exponentially I&I velocity observer for system (1) and (2). Let us define the estimation error as

$$\tilde{v}_i = \hat{v}_i - v_i, \quad \tilde{\omega}_i = \hat{\omega}_i - \omega_i, \quad \tilde{\theta}_i = \vartheta_i - \theta_i \quad (7)$$

where ϑ_i is the estimate of the angular positions for each agent.

The first two estimations are designed to contain a proportional and an integral part, as follows:

$$\begin{aligned} \dot{\hat{v}}_i &= v_{I_i} + v_{P_i}(\vartheta_i, z_i) \\ \dot{\hat{\omega}}_i &= \omega_{I_i} + \omega_{P_i}(\vartheta_i). \end{aligned} \quad (8)$$

The objective in the observer design is to find the dynamics of the integral parts v_{I_i}, ω_{I_i} and to set the proportional terms v_{P_i}, ω_{P_i} ; such that the estimation error (7) exponentially converges to zero for all initial conditions.

Proposition 1: Consider system (1) and (2), design the integral parts of the I&I velocity observer (8) and ϑ -dynamic as

$$\dot{v}_{I_i} = -k_{v_i} \varphi^\perp(\vartheta_i) z_i \dot{\vartheta}_i - k_{v_i} \varphi^\top(\vartheta_i) \varphi(\theta_i) \dot{v}_i + u_{vi} \quad (9)$$

$$\dot{\omega}_{I_i} = -k_{\omega_i} \hat{\omega}_i + u_{\omega i} \quad (10)$$

$$\dot{\tilde{\theta}}_i = -\gamma_i \left(\tilde{\theta}_i, r_{v_i} \right) \tilde{\theta}_i + \hat{\omega}_i \quad (11)$$

with proportional terms as

$$v_{P_i} = k_{v_i} \varphi^\top(\vartheta_i) z_i \quad (12)$$

$$\omega_{P_i} = k_{\omega_i} \tilde{\theta}_i \quad (13)$$

and r_{v_i} as a dynamic scaling given by

$$\dot{r}_{v_i} = -\frac{k_{v_i}}{4} \tilde{r}_{v_i} + 2k_{v_i} \tilde{\theta}_i^2 r_{v_i}, \quad r_{v_i}(0) > 1 \quad (14)$$

where $\tilde{r}_{v_i} = (r_{v_i} - 1)$ and the set $\{r \in \mathbb{R} | r \geq 1\}$ is invariant for dynamics (14), k_{ω_i} are free positive constants and k_{v_i} and γ_i are positive (possibly time varying) functions that will be defined later.

Then, for all initial conditions $(z_i(0), \theta_i(0), v_i(0), \omega_i(0)) \in \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}$, and $r_{v_i}(0) > 1$, we ensure

$$\left\| \begin{bmatrix} \tilde{\omega}_i(t) \\ \tilde{v}_i(t) \\ \tilde{\theta}_i(t) \\ \tilde{r}_{v_i}(t) \end{bmatrix} \right\| \leq \kappa_i \exp^{-\rho_i(t-t_0)} \left\| \begin{bmatrix} \tilde{\omega}_i(0) \\ \tilde{v}_i(0) \\ \tilde{\theta}_i(0) \\ \tilde{r}_{v_i}(0) \end{bmatrix} \right\| \quad (15)$$

for some constants $\kappa_i, \rho_i > 0$ (independent of t_0) and all $t \geq t_0 \geq 0$. $\triangle$

Proof: On one hand, the $\tilde{\omega}_i$ dynamics is given by¹

$$\begin{aligned} \dot{\tilde{\omega}}_i &= \dot{\omega}_i - \dot{\omega}_i \\ &= \dot{\omega}_{I_i} + \nabla \omega_{P_i}(\hat{\omega}_i - \tilde{\omega}_i) - u_{\omega_i} = -k_{\omega_i} \tilde{\omega}_i \end{aligned} \quad (16)$$

where to obtain (16), we have employed (2), (10), and (13). Proposing as Lyapunov function $H_{\omega_i} = \frac{1}{2} \tilde{\omega}_i^2$, its time derivate along the solution (16) leads to $\dot{H}_{\omega_i} = -2k_{\omega_i} H_{\omega_i}$. Thus, $\tilde{\omega}_i$ exponentially converges to zero.

On the other hand, the dynamic behavior of $\tilde{v}$ is governed by

$$\begin{aligned} \dot{\tilde{v}}_i &= \dot{v}_i - \dot{v}_i \\ &= \dot{v}_{I_i} + \nabla_{\vartheta_i} v_{P_i} \hat{v}_i + \nabla_{z_i}^\top v_{P_i} \varphi(\theta_i) (\hat{v}_i - \tilde{v}_i) - u_{1_i}. \end{aligned} \quad (17)$$

Choosing $\dot{v}_{I_i}$ as (9) we get $\dot{\tilde{v}}_i = -\nabla_{z_i}^\top v_{P_i} \varphi(\theta_i) \tilde{v}_i$.

Moreover, using the definition of v_{P_i} given by (12) yields

$$\begin{aligned} \dot{\tilde{v}}_i &= -k_{v_i} \varphi^\top(\vartheta_i) \varphi(\theta_i) \tilde{v}_i \\ &= k_{v_i} \{ \pm \varphi^\top(\theta_i) - \varphi^\top(\vartheta_i) \} \varphi(\theta_i) \tilde{v}_i \\ &= -k_{v_i} \tilde{v}_i + \Delta_{v_i}(\theta_i, \vartheta_i) \tilde{v}_i \end{aligned} \quad (18)$$

where $\Delta_{v_i}(\vartheta_i, \theta_i) = -k_{v_i} \{ \varphi^\top(\vartheta_i) - \varphi^\top(\theta_i) \} \varphi(\theta_i)$ and verifies $\Delta_{v_i}(\vartheta_i, \theta_i) = 0$ if $\tilde{\theta} = 0$. Furthermore, this mapping satisfies

$$\|\Delta_{v_i}(\vartheta_i, \theta_i)\| \leq 2k_{v_i} \tilde{\theta}_i \quad (19)$$

and plays the role of a disturbance that will be dominated with a dynamic scaling and the observer dynamics ϑ_i given by (11).

Let us define the dynamically scaled off-the-manifold coordinate

$$z_{v_i} = \frac{1}{r_{v_i}} \tilde{v}_i. \quad (20)$$

Its dynamic behavior is given by

$$\dot{z}_{v_i} = -k_{v_i} z_{v_i} + \Delta_{v_i}(\vartheta_i, \theta_i) z_{v_i} - \frac{\dot{r}_{v_i}}{r_{v_i}} z_{v_i}. \quad (21)$$

¹Given $f: \mathbb{R}^n \rightarrow \mathbb{R}$ we define $\nabla f := \left(\frac{\partial f}{\partial x} \right)^\top$, $\nabla_{x_i} f := \left(\frac{\partial f}{\partial x_i} \right)^\top$, where $x_i \in \mathbb{R}^p$ is an element of the vector x .

In addition, the dynamic $\tilde{\theta}$ corresponds to

$$\dot{\tilde{\theta}}_i = \dot{\theta}_i - (\dot{\omega}_i - \tilde{\omega}_i) := -\gamma(r_{v_i}) \tilde{\theta}_i + \tilde{\omega}_i \quad (22)$$

where to get the last equality we have used (11).

Consider the Lyapunov function candidate

$$H_{v_i} = \frac{1}{2} \left(\tilde{\omega}_i^2 + z_{v_i}^2 + \tilde{\theta}_i^2 + \tilde{r}_{v_i}^2 \right). \quad (23)$$

Taking its time derivative, we get

$$\begin{aligned} \dot{H}_{v_i} &= -k_{\omega_i} \tilde{\omega}_i^2 + z_{v_i}^2 \left(-k_{v_i} + \Delta_{v_i}(\vartheta_i, \theta_i) - \frac{\dot{r}_{v_i}}{r_{v_i}} \right) \\ &\quad + \tilde{\theta}_i \left(-\gamma(r_{v_i}) \tilde{\theta}_i + \tilde{\omega}_i \right) + \tilde{r}_{v_i} \dot{r}_{v_i} \\ &\leq -k_{\omega_i} \tilde{\omega}_i^2 + z_{v_i}^2 \left(-\frac{k_{v_i}}{2} + \frac{1}{2k_{v_i}} \|\Delta_{v_i}(\vartheta_i, \theta_i)\|^2 - \frac{\dot{r}_{v_i}}{r_{v_i}} \right) \\ &\quad - \gamma(r_{v_i}) \tilde{\theta}_i^2 + \frac{1}{2} \tilde{\theta}_i^2 + \frac{1}{2} \tilde{\omega}_i^2 + \tilde{r}_{v_i} \dot{r}_{v_i}. \end{aligned} \quad (24)$$

Hence, using inequality (19) and choosing $\dot{r}_{v_i}$ as (14), we obtain

$$\begin{aligned} \dot{H}_{v_i} &\leq -\left(k_{\omega_i} - \frac{1}{2} \right) \tilde{\omega}_i^2 - k_{v_i} z_{v_i}^2 \left(\frac{1}{2} - \frac{\tilde{r}_{v_i}}{4r_{v_i}} \right) \\ &\quad - \tilde{\theta}_i^2 \left(\gamma(r_{v_i}) - \frac{1}{2} - 2k_{v_i} \tilde{r}_{v_i} r_{v_i} \right) - \frac{k_{v_i}}{4} \tilde{r}_{v_i}^2 \\ &\leq -k_{1_i} \tilde{\omega}_i^2 - k_{2_i} \tilde{\theta}_i^2 - k_{3_i} z_{v_i}^2 - k_{3_i} \tilde{r}_{v_i}^2 \end{aligned} \quad (25)$$

where the last inequality have been obtained using the fact that $1 > \frac{\tilde{r}_{v_i}}{r_{v_i}} > 0$ and selecting

$$k_{\omega_i} = k_{1_i} + \frac{1}{2}, \quad \gamma(r_{v_i}) = k_{2_i} + \frac{1}{2} + 2k_{v_i} \tilde{r}_{v_i} r_{v_i}, \quad k_{v_i} = 4k_{3_i}. \quad (26)$$

As consequence (15) is established. $\blacksquare$

B. Leaderless Consensus-Based Formation Control

Consider the control laws

$$u_{v_i} = -p_{v_i} \varphi(\theta_i)^\top e_{z_i} - d_{v_i} \hat{v}_i \quad (27a)$$

$$u_{\omega_i} = -p_{\omega_i} e_{\theta_i} - d_{\omega_i} \hat{\omega}_i + \alpha_i(t, \theta_i, e_{z_i}) \quad (27b)$$

where

$$e_{z_i} := \sum_{j \in \mathcal{N}_i} a_{ij} [\bar{z}_i - \bar{z}_j(t - T_{ji}(t))] \quad (28)$$

$$e_{\theta_i} := \sum_{j \in \mathcal{N}_i} a_{ij} [\theta_i - \theta_j(t - T_{ji}(t))] \quad (29)$$

with $\hat{\omega}_i$ and $\hat{v}$ as the velocity estimators given by (8), the constants p_{v_i} and $p_{\omega_i} > 0$ are proportional gains, d_{v_i} and $d_{\omega_i} > 0$ are damping constant gains, and a vanishing function α_i defined as

$$\alpha_i(t, \theta_i, e_{z_i}) := k_{\alpha_i} f_i(t) \varphi(\theta_i)^\perp e_{z_i} \quad (30)$$

where $k_{\alpha_i} > 0$, $\varphi(\theta_i)^\perp = [-\sin(\theta_i) \cos(\theta_i)]^\top$ is the annihilator of φ , i.e., $\varphi(\theta_i)^\top \varphi(\theta_i)^\perp = \varphi(\theta_i)^\perp \varphi(\theta_i) = 0$. Furthermore, $f_i \in \mathcal{C}^2(\mathbb{R}_{\geq 0}, \mathbb{R})$ satisfies, by construction, $f_i, \dot{f}_i, \ddot{f}_i \in \mathcal{L}_\infty$, $\lim_{t \rightarrow \infty} f_i(t) \neq 0$, and $\lim_{t \rightarrow \infty} \dot{f}_i(t) \neq 0$. Then, we have the following.

Proposition 2: Consider system (1) and (2), under Assumptions A1 and A2, in closed loop with (27). Then, the leaderless consensus control

goal is achieved, that is, (5) and (6) hold with (z_c, θ_c) , provided that

$$d_{vi} > p_{vi} \sum_{j \in \mathcal{N}_i} a_{ij} \left(\beta_i + \frac{\bar{T}_{ij}^2}{\beta_j} \right) \quad (31)$$

$$d_{\omega i} > p_{\omega i} \sum_{j \in \mathcal{N}_i} a_{ij} \left(\varepsilon_i + \frac{\bar{T}_{ij}^2}{\varepsilon_j} \right) \quad (32)$$

for some $\beta_i, \varepsilon_i > 0$, for all $i \leq N$. $\square$

Before presenting the proof, we notice that the closed-loop equations can be rewritten in the form

$$\begin{cases} \dot{z}_i = \varphi(\theta_i) v_i \\ \dot{v}_i = -d_{vi} v_i - p_{vi} \varphi(\theta_i)^\top e_{zi} - d_{vi} r_{vi} z_{vi} \end{cases} \quad (33)$$

$$\begin{cases} \dot{\theta}_i = \omega_i \\ \dot{\omega}_i = -d_{\omega i} \omega_i - p_{\omega i} e_{\theta i} - d_{\omega i} \tilde{\omega}_i + \alpha_i(t, \theta_i, e_{zi}) \end{cases} \quad (34)$$

where to get the two-subsystems, we have used the definition of the observation errors (7) and (20).

Proof of Proposition 2: The proof relies on Barbalat's lemma and standard, yet cumbersome signal-chasing arguments. We start by showing that $v_i \rightarrow 0$ and that the solutions of Σ_v are bounded. To that end, consider the following energy-like function:

$$\mathcal{E}_i := \frac{1}{2p_{vi}} v_i^2 + \frac{1}{4} \sum_{j \in \mathcal{N}_i} a_{ij} |\bar{z}_i - \bar{z}_j|^2$$

where the first term stands for the kinetic energy and the second for the potential energy stored in the robots interconnection. $\dot{\mathcal{E}}_i$ evaluated along (33) yields

$$\begin{aligned} \dot{\mathcal{E}}_i &= -\frac{d_{vi}}{p_{vi}} v_i^2 - \dot{z}_i^\top \sum_{j \in \mathcal{N}_i} a_{ij} (\bar{z}_i - \bar{z}_j(t - T_{ji}(t))) - \frac{d_{vi}}{p_{vi}} v_i r_{vi} z_{vi} \\ &\quad + \frac{1}{2} \sum_{j \in \mathcal{N}_i} a_{ij} (\dot{z}_i - \dot{z}_j)^\top (\bar{z}_i - \bar{z}_j). \end{aligned}$$

Defining the total energy of all the N systems as $\mathcal{E} = \sum_{i \leq N} \mathcal{E}_i$ and making use of [28, Lemma 6.1] to obtain, under Assumption A1

$$\frac{1}{2} \sum_{i \leq N} \sum_{j \in \mathcal{N}_i} a_{ij} (\dot{z}_i - \dot{z}_j)^\top (\bar{z}_i - \bar{z}_j) = \sum_{i \leq N} \sum_{j \in \mathcal{N}_i} a_{ij} \dot{z}_i^\top (\bar{z}_i - \bar{z}_j)$$

then it holds that

$$\dot{\mathcal{E}} = - \sum_{i \leq N} \left[\frac{d_{vi}}{p_{vi}} (v_i^2 - v_i r_{vi} z_{vi}) + \sum_{j \in \mathcal{N}_i} a_{ij} \dot{z}_i^\top \int_{t-T_{ji}(t)}^t \dot{z}_j(\sigma) d\sigma \right]$$

where we have further employed the fact that

$$\bar{z}_j - \bar{z}_j(t - T_{ji}(t)) = \int_{t-T_{ji}(t)}^t \dot{z}_j(\sigma) d\sigma.$$

Next, we apply Young's and Cauchy-Schwarz' inequalities on the last right-hand-term of $\dot{\mathcal{E}}$ to obtain

$$\begin{aligned} -\dot{z}_i^\top \int_{t-T_{ji}}^t \dot{z}_j(\sigma) d\sigma &\leq \frac{\beta_i}{2} |\dot{z}_i|^2 + \frac{1}{2\beta_i} \left| \int_{t-T_{ji}}^t \dot{z}_j(\sigma) d\sigma \right|^2 \\ &\leq \frac{\beta_i}{2} |\dot{z}_i|^2 + \frac{\bar{T}_{ji}}{2\beta_i} \int_{t-\bar{T}_{ji}}^t |\dot{z}_j(\sigma)|^2 d\sigma \end{aligned}$$

for any $\beta_i > 0$. Since $\varphi^\top \varphi = 1$, we have $|\dot{z}_i|^2 = v_i^2$. Moreover, using Young's inequality on the term $v_i r_{vi} z_{vi}$ we can further write

$$\dot{\mathcal{E}} \leq - \sum_{i \leq N} \left[\frac{1}{2} \left(\frac{d_{vi}}{p_{vi}} - \beta_i l_{ii} \right) v_i^2 - \frac{d_{vi}}{2p_{vi}} r_{vi}^2 z_{vi}^2 \right]$$

$$+ \sum_{i \leq N} \sum_{j \in \mathcal{N}_i} a_{ij} \frac{\bar{T}_{ji}}{2\beta_i} \int_{t-\bar{T}_{ji}}^t |v_j(\sigma)|^2 d\sigma \quad (35)$$

where $l_{ii} := \sum_{j \in \mathcal{N}_i} a_{ij}$ is the i th element in the diagonal of the Laplacian matrix.

In order to eliminate the influence of the delays in the time derivative of the energy function $\mathcal{E}$, we employ the following Lyapunov-Krasovskii functional:

$$\mathcal{D} := \frac{1}{2} \sum_{i \leq N} \frac{1}{\beta_i} \sum_{j \in \mathcal{N}_i} a_{ij} \bar{T}_{ji} \int_{T_{ji}}^0 \int_{t+\sigma}^t v_j(\eta)^2 d\eta d\sigma.$$

Note that $\dot{\mathcal{D}}$ is

$$\dot{\mathcal{D}} = \frac{1}{2} \sum_{i \leq N} \frac{1}{\beta_i} \sum_{j \in \mathcal{N}_i} a_{ij} \left(\bar{T}_{ji}^2 v_j^2 - \bar{T}_{ji} \int_{t-\bar{T}_{ji}}^t v_j^2(\sigma) d\sigma \right).$$

Clearly, the last term of $\dot{\mathcal{D}}$ serves to eliminate the last term in (35).

Finally, consider the complete Lyapunov-Krasovskii functional given by

$$\mathcal{V} := \mathcal{E} + \mathcal{D} + \sum_{i \leq N} \left(\mathcal{H}_{vi} + \frac{1}{2} (\gamma_{vi} - 1) z_{vi}^2 \right) \quad (36)$$

where $\mathcal{H}_{vi}$ is defined in (23) and γ_{vi} is a free positive constant.

The total derivative of $\mathcal{V}$ along the trajectories of (16), (21), (22), and (33), yields

$$\begin{aligned} \dot{\mathcal{V}} &\leq - \sum_{i \leq N} \left[\frac{1}{2} \left(\frac{d_{vi}}{p_{vi}} - \beta_i l_{ii} \right) v_i^2 - \sum_{j \in \mathcal{N}_i} a_{ij} \frac{\bar{T}_{ji}^2}{2\beta_i} v_j^2 \right. \\ &\quad \left. - \frac{d_{vi}}{2p_{vi}} r_{vi}^2 z_{vi}^2 + k_{1i} \tilde{\omega}_i^2 + k_{2i} \tilde{\theta}_i^2 + \gamma_{vi} k_{3i} z_{vi}^2 + k_{3i} \tilde{r}_{vi}^2 \right]. \end{aligned}$$

Following [29] and after some straightforward calculations, we obtain

$$\begin{aligned} \dot{\mathcal{V}} &\leq - \frac{1}{2} \sum_{i \leq N} \left[\frac{d_{vi}}{p_{vi}} - \sum_{j \in \mathcal{N}_i} a_{ij} \left(\beta_i + \frac{\bar{T}_{ij}^2}{\beta_j} \right) \right] v_i^2 \\ &\quad - \sum_{i \leq N} \left[k_{1i} \tilde{\omega}_i^2 + k_{2i} \tilde{\theta}_i^2 + k_{z_i} z_{vi}^2 + k_{3i} \tilde{r}_{vi}^2 \right] \end{aligned}$$

with gain k_{z_i} as

$$k_{z_i} = \gamma_{vi} k_{3i} - \frac{d_{vi}}{2p_{vi}} r_{vi}^2. \quad (37)$$

Since the second term of k_{z_i} is definitive negative and r_{vi} is decrescent (time varying) with an a priori known initial condition, we can choose γ_{vi} to satisfy $\gamma_{vi} > \frac{d_{vi}}{2k_{3i} p_{vi}} r_{vi}^2(0)$ —or γ_{vi} sufficiently large—to ensure the positivity of k_{z_i} . Thus, after (31), it follows that there exists $\lambda_i > 0$ such that

$$\dot{\mathcal{V}} \leq - \sum_{i \leq N} \left(\lambda_i v_i^2 + k_{1i} \tilde{\omega}_i^2 + k_{2i} \tilde{\theta}_i^2 + k_{z_i} z_{vi}^2 + k_{3i} \tilde{r}_{vi}^2 \right).$$

This implies that $(v_i, \tilde{\omega}_i, \tilde{\theta}_i, z_{vi}, \tilde{r}_{vi}) \in \mathcal{L}_2$ and, since $\mathcal{V}$ is positive definite and radially unbounded with respect their arguments and $\bar{z}_i - \bar{z}_j$, the latter belong to $\mathcal{L}_\infty$ and, consequently, $e_{zi} \in \mathcal{L}_\infty$. Moreover, since φ is uniformly bounded, it follows from (33) that $\dot{v}_i \in \mathcal{L}_\infty$ and, therefore, $\lim_{t \rightarrow \infty} v_i(t) = 0$.

Now, we analyze the behavior of the solutions of (34) with the aim at establishing the second limit. To that end, consider the subsystem

(34) with (for the time being) $\alpha_i \equiv 0$ and the Lyapunov–Krasovskii functional

$$\mathcal{W} := \sum_{i \leq N} \left[\frac{1}{2} \tilde{\omega}_i^2 + \frac{1}{2p_{\omega i}} \omega_i^2 + \frac{1}{4} \sum_{j \in N_i} a_{ij} (\theta_i - \theta_j)^2 + \frac{1}{2\varepsilon_i} \sum_{j \in N_i} a_{ij} \bar{T}_{ji} \int_{-\bar{T}_{ji}}^0 \int_{t+\sigma}^t \omega_j^2(\eta) d\eta d\sigma \right] \quad (38)$$

where $\varepsilon_i > 0$. Proceeding as we did previously for the differentiation of $\mathcal{V}$, we obtain

$$\dot{\mathcal{W}} \leq - \sum_{i \leq N} \left[\frac{1}{2} \left(\frac{d_{\omega i}}{p_{\omega i}} - \varepsilon_i l_{ii} \right) \omega_i^2 - \frac{d_{\omega i}}{2p_{\omega i}} \tilde{\omega}_i^2 + k_{\omega i} \tilde{\omega}_i^2 - \sum_{j \in N_i} a_{ij} \frac{\bar{T}_{ji}^2}{2\varepsilon_i} \omega_j^2 \right].$$

From (26), we have that $k_{\omega i} = k_{1i} + \frac{1}{2}$, then defining $k_{1i} = \gamma_{\omega i} + \frac{d_{\omega i}}{2p_{\omega i}}$ with free positive gain $\gamma_{\omega i}$, we get

$$\dot{\mathcal{W}} \leq - \frac{1}{2} \sum_{i \leq N} \left(\frac{d_{\omega i}}{p_{\omega i}} - \sum_{j \in N_i} a_{ij} \left(\varepsilon_i + \frac{\bar{T}_{ij}}{\varepsilon_j} \right) \right) \omega_i^2 - \sum_{i \leq N} \bar{\gamma}_{\omega i} \tilde{\omega}_i^2$$

where $\bar{\gamma}_{\omega i} = \gamma_{\omega i} + \frac{1}{2}$. Therefore, after (32), it follows that there exists $\lambda_i > 0$ such that $\dot{\mathcal{W}} \leq - \sum_{i \leq N} (\lambda_i \omega_i^2 + \bar{\gamma}_{\omega i} \tilde{\omega}_i^2)$. Since $\mathcal{W}$ is positive definite and radially unbounded with regards to ω_i and $\theta_i - \theta_j$, it follows that $(\omega_i, \tilde{\omega}_i) \in \mathcal{L}_2 \cap \mathcal{L}_\infty$ and $\theta_i - \theta_j \in \mathcal{L}_\infty$. This, in turn, implies that $\dot{\omega}_i \in \mathcal{L}_\infty$ so, after Barbalat Lemma, $\lim_{t \rightarrow \infty} \omega_i(t) = 0$.

We proceed now to establish the limit in (6). To that end, observe that, since $\lim_{t \rightarrow \infty} \int_0^t \dot{\omega}_i(\sigma) d\sigma = \lim_{t \rightarrow \infty} \omega_i(t) - \omega_i(0) = \omega_i(0)$, it follows, from Barbalat lemma that $\lim_{t \rightarrow \infty} \dot{\omega}_i(t) = 0$ if $\dot{\omega}_i$ is uniformly continuous. The latter holds since

$$\dot{\omega}_i = -[p_{\omega i} \dot{e}_{\theta i} + d_{\omega i} \dot{\omega}_i - d_{\omega i} k_{\omega i} \tilde{\omega}_i]$$

is bounded; indeed, this follows from Assumption A2 and the fact that $\tilde{\omega}_i, \dot{\omega}_i, \omega_i \in \mathcal{L}_\infty$.

Now, from the second equation in (34), restricted to $\alpha_i \equiv 0$, it follows that $\lim_{t \rightarrow \infty} (\dot{\omega}_i(t), \omega_i(t), \tilde{\omega}_i(t)) = 0$ imply that $\lim_{t \rightarrow \infty} e_{\theta i}(t) = 0$ or, equivalently that

$$\lim_{t \rightarrow \infty} \sum_{j \in N_i} a_{ij} \left(\theta_i(t) - \theta_j(t) + \int_{t-T_{ji}(t)}^t \omega_j(\sigma) d\sigma \right) = 0.$$

Then, since $\omega_i \rightarrow 0$ and $T_{ji}(t)$ is globally bounded (see A2), we have that $\lim_{t \rightarrow \infty} \sum_{j \in N_i} a_{ij} [\theta_i(t) - \theta_j(t)] = 0$, which implies that $\lim_{t \rightarrow \infty} L\theta(t) = 0$, where $\theta = \text{col}(\theta_i) \in \mathbb{R}^N$. Thus

$$\lim_{t \rightarrow \infty} \theta_i(t) = \theta_c. \quad (39)$$

All the previous arguments hold provided that $\alpha_i \equiv 0$. If this is not the case, we remark that since, f_i , e_{zi} , and $\varphi^\perp$ are bounded along trajectories, so is α_i —see (30). This and the fact that Σ_ω is a marginally stable linear time-varying system with uniformly bounded time delays ensure that $\tilde{\omega}_i, \dot{\omega}_i, \omega_i \in \mathcal{L}_\infty$, by [30, Proposition 3] and so is $e_{\theta i}$.

The rest of the proof mostly consists in establishing that $e_{zi} \rightarrow 0$, which, since $v_i \rightarrow 0$, implies that $\tilde{z}_i \rightarrow \bar{z}_c$, thereby completing (6). To see the latter, since $\dot{\tilde{z}}_i = \varphi(\theta_i) v_i$, note that

$$e_{zi} = \sum_{j \in N_i} a_{ij} \left(\tilde{z}_i - \tilde{z}_j + \int_{t-T_{ji}(t)}^t \varphi(\theta_j(\sigma)) v_j(\sigma) d\sigma \right)$$

so, defining $e_z := [e_{z1} \cdots e_{zN}]^\top$ and $\tilde{z} := [\tilde{z}_{11} \cdots \tilde{z}_{iN}]$, respectively, we obtain

$$e_z = (L \otimes I_2) \tilde{z} + \text{col} \left(\sum_{j \in N_i} a_{ij} \int_{t-T_{ji}(t)}^t \varphi(\theta_j(\sigma)) v_j(\sigma) d\sigma \right).$$

Therefore, $e_{zi} = 0$ and $v_i = 0$ imply that $(L \otimes I_2) \tilde{z} = 0$ which, in view of the properties of L , implies the existence of $\tilde{z}_c \in \mathbb{R}^2$ such that $\tilde{z} = 1_N \otimes \tilde{z}_c$, or $\tilde{z}_i(t) = \tilde{z}_c$ for all $i \leq n$.

We proceed now to show that $e_{zi} \rightarrow 0$. To that end, we stress that, since $\lim_{t \rightarrow \infty} v_i(t) = 0$

$$\lim_{t \rightarrow \infty} \int_0^t \dot{v}_i(\sigma) d\sigma = \lim_{t \rightarrow \infty} v_i(t) - v_i(0) = v_i(0).$$

Moreover, in view of Assumption A2 and the fact that $\theta_i, \omega_i, \dot{\omega}_i, v_i, \dot{v}_i, r_{v_i}, z_{v_i}$, and $e_{zi} \in \mathcal{L}_\infty$, so does

$$\begin{aligned} \ddot{v}_i &= -p_{vi} \left[\omega_i \varphi^{\perp\top} e_{zi} + \varphi^\top \dot{e}_{zi} + \frac{d_{vi}}{p_{vi}} \dot{v}_i + \frac{d_{vi}}{p_{vi}} \overbrace{(r_{v_i} z_{v_i})} \right] \\ &= -p_{vi} \left[\omega_i \varphi^{\perp\top} e_{zi} + \varphi^\top \dot{e}_{zi} + \frac{d_{vi}}{p_{vi}} \dot{v}_i \right] \\ &\quad - d_{vi} z_{v_i} r_{v_i} (\Delta_{v_i} - k_{v_i}) \end{aligned} \quad (40)$$

where to get the last term of the second equality we have used (21).

After Barbalat lemma, it follows that $\lim_{t \rightarrow \infty} \dot{v}_i(t) = 0$ and, in turn after (33), it follows that $\lim_{t \rightarrow \infty} \varphi(\theta(t))^\top e_{zi}(t) = 0$.

Note, however, that the solutions to $\varphi(\theta)^\top e_{zi} = 0$ are of the form $e_{zi} = c\varphi^\perp$. Notwithstanding, so are the solutions to $\varphi(\theta)^\perp e_{zi} = 0$ on the right-hand side of (40). Therefore, the unique solution to both equations is $c = 0$ or, equivalently, $e_{zi} = 0$, since $\varphi^{\perp\top} \varphi^\perp = 1$.

Thus, it is left to establish that $\varphi(\theta(t))^\perp e_{zi}(t) \rightarrow 0$. To that end, we use once more Barbalat lemma to show that $\ddot{v}_i \rightarrow 0$. First, note that $\lim_{t \rightarrow \infty} \int_0^t \ddot{v}_i(\sigma) d\sigma = \lim_{t \rightarrow \infty} \dot{v}_i(t) - \dot{v}_i(0) = -\dot{v}_i(0)$. Then, to show that $\ddot{v}_i$ is uniformly continuous, we use (40) and $\dot{\varphi}^\perp = -\omega_i \varphi$ to obtain

$$\begin{aligned} \ddot{v}_i &= -p_{vi} \left[\dot{\omega}_i \varphi^{\perp\top} e_{zi} + 2\omega_i \varphi^{\perp\top} \dot{e}_{zi} - \omega_i^2 \varphi^\top e_{zi} + \varphi^\top \ddot{e}_{zi} \right. \\ &\quad \left. + \frac{d_{vi}}{p_{vi}} \ddot{v}_i + \frac{d_{vi}}{p_{vi}} \overbrace{(r_{v_i} z_{v_i})} \right] \\ &= -p_{vi} \left[\dot{\omega}_i \varphi^{\perp\top} e_{zi} + 2\omega_i \varphi^{\perp\top} \dot{e}_{zi} - \omega_i^2 \varphi^\top e_{zi} + \varphi^\top \ddot{e}_{zi} \right. \\ &\quad \left. + \frac{d_{vi}}{p_{vi}} \ddot{v}_i \right] - d_{vi} r_{v_i} z_{v_i} \left\{ \dot{\Delta}_{v_i} + (\Delta_{v_i} - k_{v_i})^2 \right\} \end{aligned} \quad (41)$$

which is bounded, in view of Assumption A2 and the boundedness of all functions on the right-hand side including $\dot{\Delta}$ (see the Appendix). This implies that $\lim_{t \rightarrow \infty} \ddot{v}_i(t) = 0$. Similar arguments apply to establish that $\lim_{t \rightarrow \infty} \ddot{v}_i(t) = 0$ and that $\lim_{t \rightarrow \infty} v_i^{(4)}(t) = 0$ where

$$\begin{aligned} v_i^{(4)} &= -p_{vi} \left[\ddot{\omega}_i \varphi^{\perp\top} e_{zi} + \dot{\omega}_i \varphi^{\perp\top} \dot{e}_{zi} - \omega_i^3 \varphi^{\perp\top} e_{zi} + \varphi^\top \ddot{\ddot{e}}_{zi} \right. \\ &\quad \left. + 3\omega_i (\varphi^{\perp\top} \ddot{e}_{zi} + \dot{\omega}_i \varphi^\top e_{zi} + \omega_i \varphi^\top \dot{e}_{zi}) + \frac{d_{vi}}{p_{vi}} \ddot{\ddot{v}}_i \right] \\ &\quad - d_{vi} \overbrace{(r_{v_i} z_{v_i})} \end{aligned} \quad (42)$$

with

$$\overbrace{(r_{v_i} z_{v_i})} = r_{v_i} z_{v_i} \left[(\Delta_{v_i} - k_{v_i}) (3\dot{\Delta}_{v_i} + (\Delta_{v_i} - k_{v_i})^2 + \ddot{\Delta}_{v_i}) \right].$$

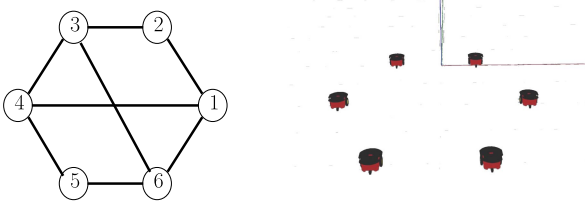


Fig. 1. (Left-hand side) Communication topology: undirected connected graph. (Right-hand side) Final configuration in the Gazebo simulator.

Therefore, since r_{v_i} converges to 1 and $z_{v_i}, v_i, \dot{v}_i, \ddot{v}_i, \ddot{v}_i^{(4)},$ and $\varphi^\top e_{z_i}$ converge to zero, so do $\dot{e}_{z_i}, \ddot{e}_{z_i}, \ddot{e}_{z_i}, \Delta_{v_i}, \dot{\Delta}_{v_i},$ and $\ddot{\Delta}_{v_i}$ (for the last ones see the Appendix). It follows that all the terms in (40), (41), and (42) converge to zero. Thusly, from (40) it follows that $\omega_i \varphi^{\perp\top} e_{z_i} \rightarrow 0$, from (41) we have $\dot{\omega}_i \varphi^{\perp\top} e_{z_i} \rightarrow 0$, and (42) implies that $\ddot{\omega}_i \varphi^{\perp\top} e_{z_i} \rightarrow 0$. Next, we use

$$\ddot{\omega}_i = -k_{\alpha i} \left[\frac{p_{\omega i}}{k_{\alpha i}} \dot{e}_{\theta i} + \frac{d_{\omega i}}{k_{\alpha i}} \dot{\omega}_i + f_i \omega_i \varphi^\top e_{z_i} - f_i \varphi^{\perp\top} \dot{e}_{z_i} - \dot{f}_i \varphi^{\perp\top} e_{z_i} - \frac{d_{\omega i}}{k_{\alpha i}} k_{\omega i} \tilde{\omega}_i \right]$$

in the first right-hand side term of (42) to conclude that the term $p_{\omega i} \dot{e}_{\theta i} \varphi^{\perp\top} e_{z_i} - \dot{f}_i (\varphi^{\perp\top} e_{z_i})^2$ converges to zero. Since, on one hand $\omega_i \varphi^{\perp\top} e_{z_i}$ converges to zero and, on the other hand, $e_{z_i} = c\varphi^\perp$ and $\varphi^{\perp\top} \varphi^\perp = 1$, it follows that ω_i also converges to zero. In turn, we also have $\dot{e}_{\theta i} \varphi^{\perp\top} e_{z_i} \rightarrow 0$. Finally, the design of f_i guarantees that $\dot{f}_i$ does not vanish, so we conclude that $\lim_{t \rightarrow \infty} \varphi^{\perp\top} (\theta_i(t)) e_{z_i}(t) = 0$. Therefore α_i vanishes and the limits in (5) holds. $\square \square \square$

IV. SIMULATION RESULTS

In order to evaluate the performance of the proposed controller, we used the simulator Gazebo and the ROS to mimic the conditions of a laboratory experimental benchmark. For each agent, we used the model of a PIONEER 3-DX wheeled robot [31], available in Gazebo's library. In the simulated scenario, it is assumed that all the robots have the same inertial and geometrical parameters given by $m = 5.64$ kg, $I = 3.115$ kg $\cdot$ m², $r = 0.09$ m, and $R = 0.157$ m.

It is important to remark that although the motion of the PIONEER 3-DX can be represented by the dynamic Equations (1) and (2), its center of mass is not located on the axis joining the two wheels' centers as it is usually assumed in the literature and in the previous sections. This means that, in right-hand sides of (1) and (2), there should appear some Coriolis terms that are quadratic in the velocities. However, as would be the case in an actual experimentation set-up, these terms constitute dynamic effects, which are not considered in the controller design. Akin to Assumption A1, we consider six PIONEER 3D-X robots communicating over the undirected connected graph illustrated in the left-hand side of Fig. 1. The initial conditions of the vehicles as well as the offsets $\delta_i = (\delta_{x_i}, \delta_{y_i})$ that determine the desired formation are given in Table I.

The time delays $T_{ji}(t)$ are generated randomly (so they are different for each pair of robots) following a normal distribution with mean $\mu = 0.3$, variance $\sigma^2 = 0.0003$ and a sample time of 10 ms—see Fig. 2 for the illustration of one of such delays. Such piece-wise constant time-varying delays are nonsmooth and do not satisfy Assumption A2, but they better represent the delays encountered in a real physical set-up. Nonetheless, despite the technical Assumption A2 not holding, full

TABLE I
INITIAL CONDITIONS

Index	x_i [m]	y_i [m]	θ_i [rad]	δ_{x_i} [m]	δ_{y_i} [m]
1	8	7	1.57	2	0
2	2	13	0.0	1	2
3	2	9	-0.39	-1	2
4	-2	6	0.39	-2	0
5	1	2	-0.39	-1	-2
6	4	4	-0.39	1	-2

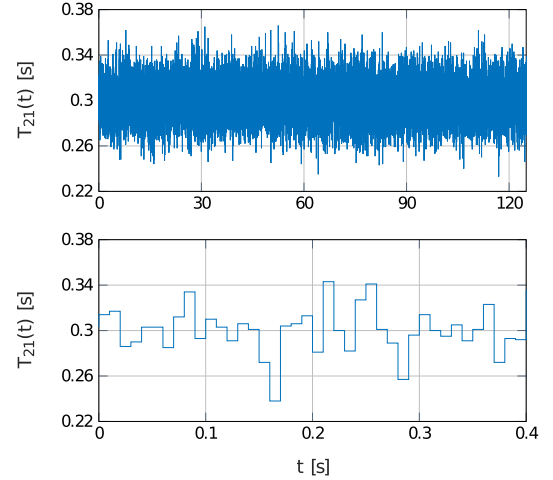


Fig. 2. Variable delay between the robot 1 and the received information from neighbor 2.

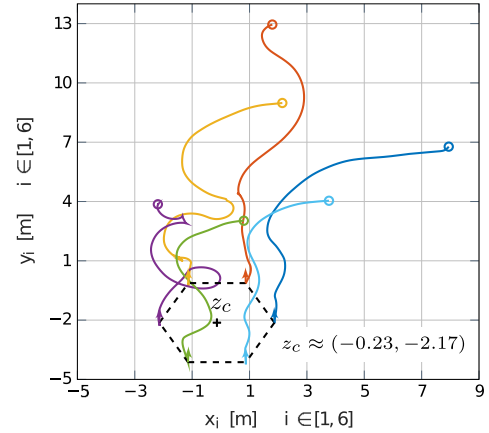


Fig. 3. Paths of the PIONEER 3D-X robots up to full consensus. The dashed lines represent the final formation centered at z_c with equal orientations illustrated by the colored arrows.

consensus is achieved, modulo a small error generated by the neglected dynamics and other realistic conditions considered in Gazebo.

The control gains were set to $d_{v_i} := d_{\omega_i} = 4$, $p_{v_i} := p_{\omega_i} = 0.3$, for all $i \in [1, 6]$. The observer gains, in turn, were selected as in (26) with $k_{1_i} = 0.5$, $k_{2_i} = 0.1$, and $k_{3_i} = 0.05$. The vanishing time-varying functions α_i , for all $i \in [1, 6]$, were taken as in (30) with $k_{\alpha i} = 0.4$ and, the (multi)periodic functions

$$f_i(t) = 2.5 + \sin(2\pi t) + 0.3 \cos(6\pi t) - 0.5 \sin(8\pi t) - 0.1 \cos(10\pi t) + \sin(\pi t) \quad \forall i \leq 6. \quad (43)$$

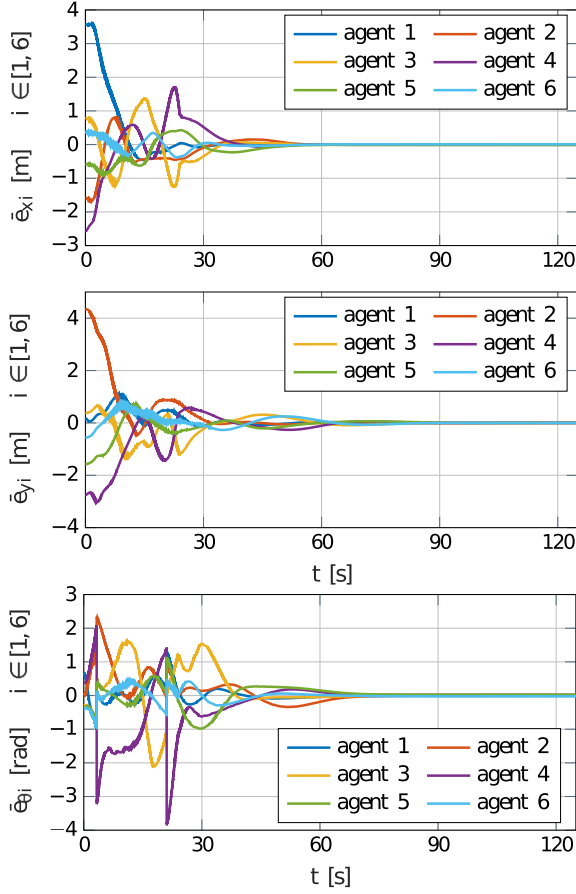


Fig. 4. Position ($\bar{e}_{xi}$, $\bar{e}_{yi}$) and orientation ($\bar{e}_{\theta i}$) consensus errors.

Furthermore, we choose $r_{v_i}(0) = 1.2$ for all agents. It can be appreciated in Fig. 3 that the vehicles achieve full consensus, see also the right-hand side of Fig. 1, where we show a screenshot of the final configuration in the Gazebo simulator. Now, to better illustrate the latter, we define the following synchronization errors:

$$\bar{e}_{zi} := \bar{z}_i - \frac{1}{N} \sum_{j \in N_i} \bar{z}_j, \quad \bar{e}_{\theta i} := \theta_i - \frac{1}{N} \sum_{j \in N_i} \theta_j. \quad (44)$$

Equation (44) represents the difference between each robot's variables and the corresponding average. In other words, the consensus objective (5) holds if the errors $\bar{e}_{zi}(t) = [\bar{e}_{xi}, \bar{e}_{yi}]$ and $\bar{e}_{\theta i}(t)$ converge to zero. We stress that $\bar{e}_{zi}(t)$ and $\bar{e}_{\theta i}(t)$ are used here only for the purpose of illustration and are not actually used in the control laws. Moreover, the linear and angular velocities of each robot are estimated correctly as it can be appreciated in Fig. 5 where the observer errors (7) are shown to converge to zero.

Although in Figs. 3 and 4, the agents achieve full consensus, it seems fit to note that from Fig. 4, the errors $\bar{e}_{\theta i}(t)$ converge to a very small neighborhood of zero. This is due to unmodeled effects that are present in the Gazebo simulator and that are not considered in dynamics (1) and (2). This smaller error induces, after a transient, a considerably small torque, which persists because it is insufficient to move the robot due to its inertia and its friction forces, this behavior can be appreciated in Fig. 6. Nevertheless, even in the presence of unmodeled dynamics introduced by the realistic simulator Gazebo and the nonsmooth time delays, under the proposed control law the robots successfully reach the desired formation, which shows its robustness.

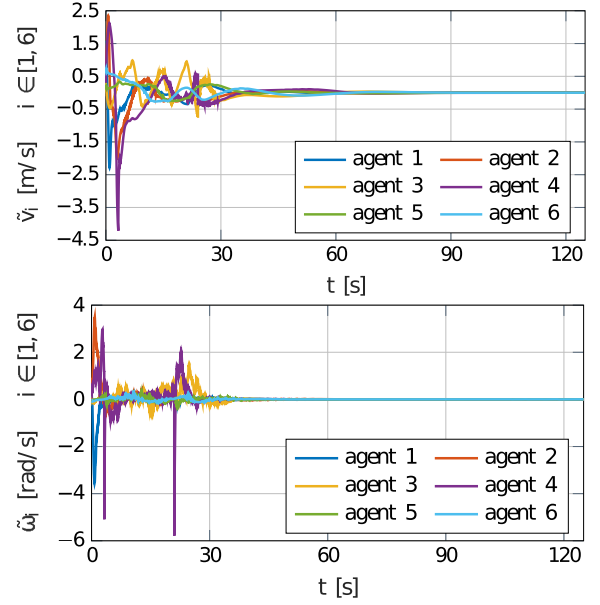


Fig. 5. Estimation errors of the linear and angular velocities.

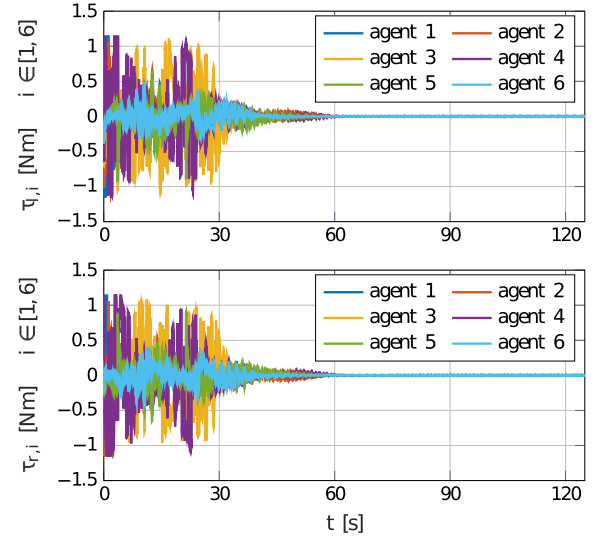


Fig. 6. Input torques.

V. CONCLUSION

In this article, we proposed a simple distributed control law, which solves the full-consensus-based formation problem for nonholonomic mobile robots under static, undirected topologies. The main contribution was that our design accounts for the complete vehicle model, time-varying communication delays, and the unavailability of velocity information. A particular advantage with respect to other laws is the implementable form of the controller that is a combination of a simple proportional-plus-damping scheme and a simple velocity observer. Furthermore, realistic simulations in the Gazebo-ROS environment showed that our law induces increased robustness with respect to measurement noise, nonsmooth delays, and uncertain dynamics that were not considered in the design steps. Future research will consider the effects of input disturbances [32], input saturations, time-varying interconnection topologies, and finite-time convergence.

APPENDIX

BOUNDEDNESS OF THE SIGNALS Δ_{v_i} , $\dot{\Delta}_{v_i}$, AND $\ddot{\Delta}_{v_i}$

Recalling the definition of Δ_{v_i}

$$\Delta_{v_i}(\vartheta_i, \theta_i) = -k_{v_i} \{ \varphi^\top(\vartheta_i) - \varphi^\top(\theta_i) \} \varphi(\theta_i)$$

we see that it is bounded by construction. Moreover, from the fact that $\bar{\theta} \rightarrow 0$ we verify $\Delta_{v_i}(\vartheta_i, \theta_i) = 0$. However, to establish this claim, we invoke Barbalat's Lemma, that is, we first prove the boundedness of $\dot{\Delta}_{v_i}$, thus, taking the time derivative of Δ leads us to

$$\begin{aligned} \dot{\Delta}_{v_i} = & -k_{v_i} [\varphi^\top(\vartheta_i) \varphi_i^{\top\perp}(\theta_i) \omega_i + \varphi^\top(\theta_i) \varphi_i^{\top\perp}(\vartheta_i) (\tilde{\omega}_i + \omega_i)] \\ & + k_{v_i} \varphi^\top(\theta_i) \varphi_i^{\top\perp}(\vartheta_i) \gamma_i(r_{v_i}) \tilde{\theta}_i \end{aligned} \quad (45)$$

with $\gamma_i(r_{v_i})$ given in (26). Since all signals on the right-hand side of (45) are bounded, we conclude that $\dot{\Delta}_{v_i}$ is also bounded. From Barbalat's Lemma, we ensure that $\Delta_{v_i} \rightarrow 0$. Mimicking the abovementioned procedure, we have that

$$\begin{aligned} \ddot{\Delta}_{v_i} = & k_{v_i} \varphi^\top(\theta_i) \varphi_i^{\top\perp}(\vartheta_i) \left\{ \gamma_i(r_{v_i}) (-\gamma_i(r_{v_i}) \tilde{\theta}_i + \tilde{\omega}_i) - k_{\omega_i} \tilde{\omega}_i \right. \\ & + 2k_{v_i} \tilde{\theta}_i \left[(\tilde{r}_{v_i} + r_{v_i}) \left(2k_{v_i} r_{v_i} \tilde{\theta}_i^2 - \frac{k_{v_i} \tilde{r}_{v_i}}{4} \right) \right] \left. \right\} \\ & + k_{v_i} \varphi^\top(\vartheta_i) \varphi(\theta_i) \left(\tilde{\omega}_i - \gamma_i(r_{v_i}) \tilde{\theta}_i \right)^2 \end{aligned} \quad (46)$$

where the all terms on the right side are bounded. Once again, using the Barbalat's Lemma, we have that (46) is bounded and that $\dot{\Delta}_{v_i} \rightarrow 0$. Finally, after simple calculations, we can ensure that $\ddot{\Delta}_{v_i}$ also converge to zero.

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